

Alex Lee

Waterloo, Canada | a55lee@uwaterloo.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Waterloo

Bachelor's in Data Science

Sep. 2023 - Apr. 2028

Waterloo, ON

Relevant Coursework: Algorithms, Data Structures, OOP, Statistics, Linear Models

SKILLS

Languages: Python, C/C++, Java, SQL(MySQL, PostgreSQL, MongoDB), R, Javascript, Typescript, Bash

ML Libraries: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, OpenCV

Cloud & DevOps: AWS(EC2, SQS), GCP(Dataflow, Compute Engine, Storage), Git, Docker, Linux

Frameworks: React, Next.js, Node.js, Express, Spring Boot, RESTful APIs, HTML5/CSS3, SASS, Bootstrap

EXPERIENCE

AI Engineer | [\[Slides\]](#)

Jan. 2026 - Aug. 2026

City of Edmonton - Future Cities Institute

Edmonton, AB

- Saved **\$500K+/year** in operational costs by replacing physical traffic sensors with an event-driven **RT-DETR** and **OpenCV** pipeline on **GCP Dataflow**, processing **1,440+ hours of traffic video**.
- Reduced web dashboard load latency by **28%** and improved concurrent user scalability by migrating the tech stack from **R Shiny** to **React** and **FastAPI**.
- Decreased cloud compute resource consumption by **40%** by implementing a **background subtraction** module using frame differencing to omit parked cars from the study.
- Lowered vehicle speed estimation error to **< 2 km/h** by calibrating camera parameters to project perspectives into **Bird's Eye View** for **3D reconstruction**.
- Containerized the entire pipeline using **Docker** and deployed to **GCP Compute Engine**, ensuring environment consistency across development and production stages.

Software Engineer

May. 2025 - Aug. 2025

Association of Korean Canadian Scientists and Engineers

Waterloo, ON

- Led the **team of 6** to develop the official AKCSE website using **Node.js**, **Express** and **MongoDB**, serving academic and career resources to **100+** students.
- Built **JWT** authentication and **role-based access control**, and improved workflow with dynamic **CRUD APIs** for news, events, and team data.

PROJECTS

Computer Vision Football Analytics | *Python, OpenCV, Pandas* | [GitHub](#) | [Link](#)

- Developed a football analysis system using **YOLOv11** and fine-tuned with **10,000+** images to detect players, referees, and the ball, achieving **0.877 mAP50** with 35.7 ms per-frame latency.
- Implemented **K-means clustering** to automatically assign players to their team, according to their jersey colors.
- Enhanced players' metrics accuracy by **27%** by engineering **camera movement compensation** feature using **Optical Flow** to track metrics consistently even on camera moving environment.
- Architected a full-stack platform using **Next.js** with an asynchronous **AWS SQS** and **EC2** architecture for video upload and real-time inference streaming.

Academic Management Web App | *Spring Boot, Javascript, MySQL, AWS EC2* | [GitHub](#)

- Built a full-stack academic management web app of **20+** pages with **50+** users using **Spring** and **MySQL**.
- Deployed and managed project on **AWS EC2** with **Apache Tomcat**, with optimized database schema and RESTful APIs.
- Implemented **10+** essential features, **CRUD** operations of course postings and assignments, self implemented real-time chat, custom course search filters, and dynamic operations and views with the database.
- Reduced query response time by **30%** by refactoring complex database queries and optimizing relational schema indexing.

Personal Portfolio Website | *Typescript, React, SASS* | [GitHub](#) | [Link](#)

- Developed a personal portfolio website using **Typescript** and **React** to showcase professional skills and projects.
- Designed a responsive UI with **SASS**, using layout and structure optimized for readability on desktop and mobile.